PD Group:	Name:

ANDERSON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS 2008

BIOLOGY **9747/02**
Higher 2

Paper 2 Core Paper

12 September 2008 Frid

2 hour

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.
Write your answers in the spaces provided on the question paper.
All working for numerical answers must be shown.

Section B

Answer any **one** question.
Write your answers on separate writing paper.

At the end of the examination,

1. fasten all your work securely together;
2. hand in your answers for Section A and Section B separately.

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The number of marks is given in brackets [ ] at the end of each question or part question.

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Section **A**

Answer **all** the questions in this section.

QUESTION 1

Fig. 1.1 is an electron micrograph of a fibroblast. Fibroblasts are flattened, irregular-shaped connective tissue cells that secrete components of extracellular matrix such as hyaluronic acid and collagen.

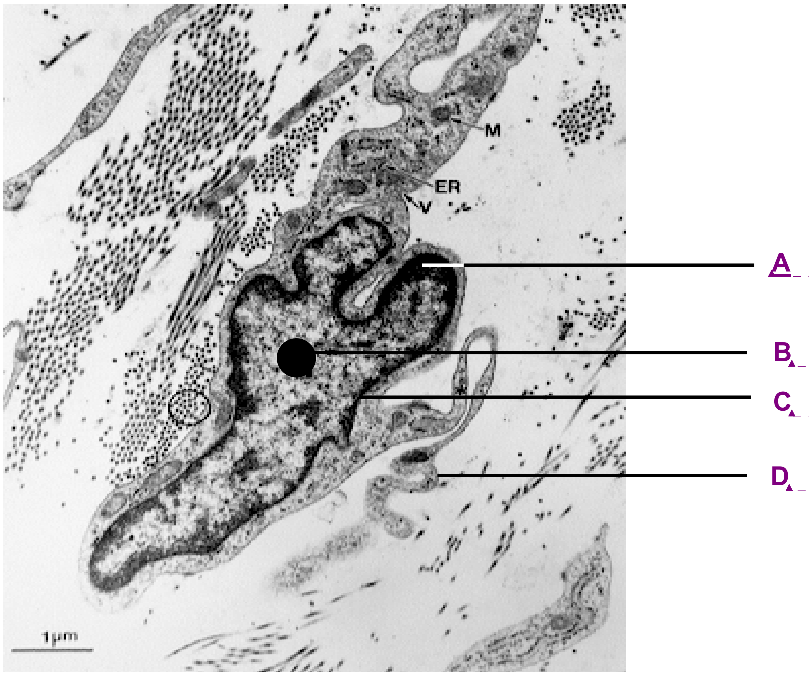

Fig. 1.1

Picture adapted from www.udel.edu/.../histopage

(a) Name structures **A** to **D**.

A Heterochromatin;
B Nucleolus;
C Nuclear envelope/ membrane;
D Cell Membrane/ Plasma membrane; [2]

(b) Explain why collagen is described as a fibrous protein.

- **Insoluble in water;**
- **Secondary structure (helical);**
- **Polypeptide chains cross link to form long fibres;**
- **Regular amino acid sequence / every 3rd residue is a glycine;** [2]

Collagen is continuously broken down in the extracellular matrix by the enzyme collagenase, which catalyses the hydrolysis of the peptide bond between the amino acids glycine and isoleucine.

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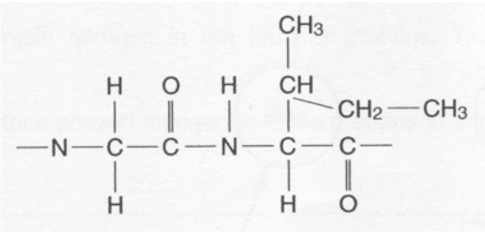
(c) Suggest how collagenase is only able to act on the peptide bond between glycine and isoleucine and not on peptide bonds between any other amino acids.

- **active site;**
- **specific complementary shape;**
- **only accepts/recognizes R grps of these 2 amino acids;** [2]

(d) Draw a diagram below to show how the peptide bond between glycine and isoleucine is broken down by hydrolysis including the product or products.



- **Correct bond broken (between C-N);**
- **Involvement of water molecules in breaking the peptide bond shown clearly;**
- **Two amino acids with free grps as follows**
  -COOH/-COO^- and –NH_2/-NH_3^+; [3]

[1 mark each]

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(c) Explain why collagen is described as a fibrous protein. [2]
<#>insoluble nature of collagen;
<#>secondary structure – three polypeptides chains are cross-linked via hydrogen bonding to each other to form a triple helix;
<#>Fibrils like parallel in bundles to form a collagen fiber;

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QUESTION 2

Huntington's Disease (HD) is a neurodegenerative disorder due to an autosomal dominant allele The disease is characterized by progressive motor dysfunction and intellectual deterioration but its symptoms usually occur when individuals are in their 30's-50. To circumvent the problem, RFLP has been used these days for early detection of the disease.

Patients with HD are known to have RFLP 1 and the diagram below shows a pair of homologous chromosomes from a β-male.

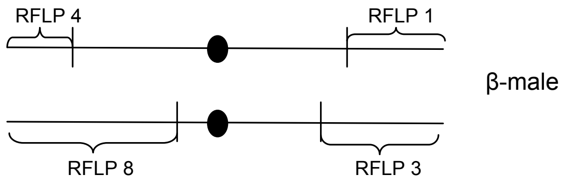


Fig. 2.1

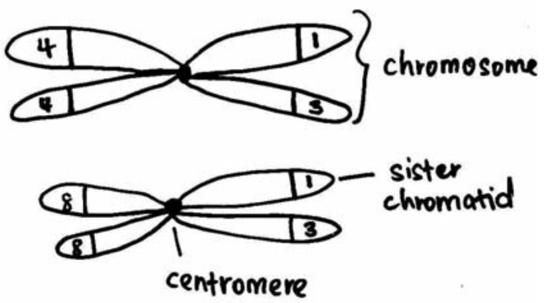
An offspring of β-male has a chromosome with RFLPs 1 and 8, contributed by the father.

(a)(i) State the event that must occur during meiosis in order to obtain a chromosome with RFLPs 1 and 8.

- **crossing over;**
- **at prophase I;**

[1]

(ii) In the space below, draw the pair of homologous chromosomes after the event that has occurred in (a)(i)



[3]

(iii) Suggest a reason why Huntington's Disease is likely to persist in populations

- **Because of its late onset;**
- **most individuals have children before they realize they have inherited the disease;**

[1]

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The micrograph below shows a lily cell undergoing cell division.

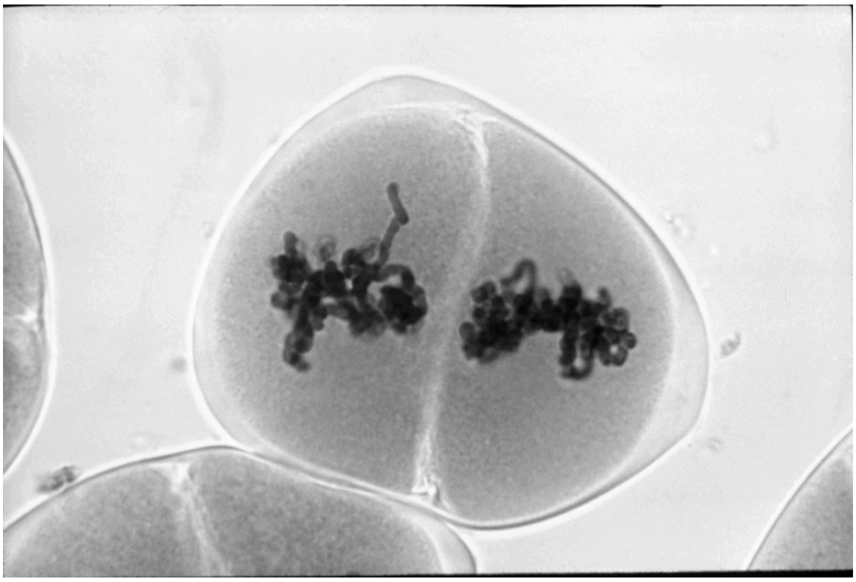

Fig.2.2 [© Clare Hasenkampf/BPS.]

(b) In reference to the micrograph above, identify:

(i) The type and stage of nuclear division

- **Meiosis;**
- **Metaphase 2;** [2]

(ii) Briefly explain the reason for your answer in (b) (i).

- **metaphase because chromosomes are aligned at equator;**
- **2 metaphase plate has formed → meiosis;**
- **the daughter cell have chromosomes aligned at the new equatorial plate which is at right angles to the first;** [2]

(c) Describe the roles of spindle fibres in meiosis.

- **pull sister chromatids/homologous chromosomes to opposite poles at anaphase;**
- **help align chromosomes at equatorial plate;** [2]

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QUESTION 3

An experiment was conducted to examine the effects of glucose and lactose on the levels of β-galactosidase in *E.coli*. Lactose and glucose were added to a culture of bacteria at the start of the experiment and the levels of each were measured at specific time intervals. The results are shown in Fig. 3.1 below.

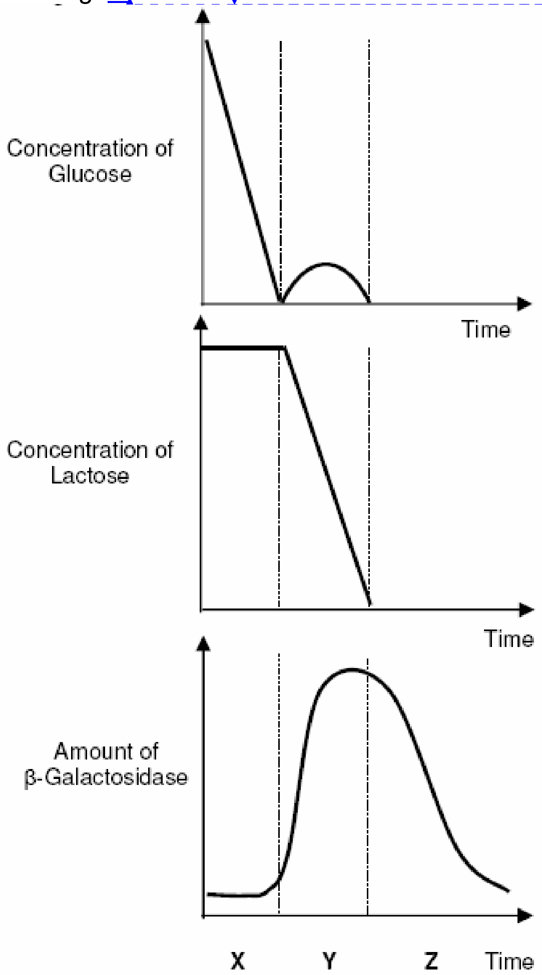

Fig. 3.1

(a) State the two mechanisms of regulation of lac operon.

- **negative control by the lac repressor;**
- **positive control by cAMP receptor protein;** [2]

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environment, typically to the presence or absence of nutrients. Some bacterial genes¶
encode for regulatory proteins which controls the expression of other genes or operons.¶
For example, *Escherichia coli* can utilize any sugar as its source of carbon and energy.¶
However, when grown in a medium containing both glucose and lactose, the bacteria will¶
use up one before the other. This preferential use of carbon source is said to be governed¶
by the "**dual control of the *lac* operon**".¶
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(b) With reference to Fig. 3.1, explain how expression levels of β -galactosidase is regulated by the lac repressor from time X to Y.

X:

- **Both glucose and lactose are present and only glucose is used;**
- **In the presence of lactose, allolactose binds to lac repressor;**
- **Thus, the Lac repressor is in its inactive configuration and does not bind to the operator;**
- **But another mechanism prevents transcription of the lac operon due to catabolite repression (ie. CAP is inactive and disengages from CAP binding site);**
- **RNA polymerase is unable to bind to the promoter/Low β –galactosidase mRNA is produced;**

Y:

- **Low glucose conc. results in high activity of adenyl cyclase (AC) which converts ATP to cAMP;**
- **This then results in high amount of cAMP-CAP complex and the complex binds upstream of *lac* promoter;**
- **Binding of cAMP-CAP enhances binding of RNA polymerase for gene transcription;**

OR

- **Once the glucose concentration has depleted, *E. coli* uses Lactose;**
- **Allolactose binds to the lac repressor;**
- **Lac repressor is inactive and does not bind to the operator;**
- **RNA polymerase is able to bind to the promoter to transcribe the mRNA of the lacoperon / Lac operon is switched on / High β-galactosidase mRNA is produced;**

[6]

(c) During period Z, suggest one mechanism that causes the amount of β-galactosidase to decrease.

- **mRNA degradation/ protein is degraded;**

OR

- **RNAi - Specific proteins are guided by the siRNA to the targeted mRNA, where they "cleave" the mRNA and it cannot be translated into protein;** [1]

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(d) In a separate experiment, *E.coli* were infected with T2 phage. Fig.3.2 shows the results of an experiment in which T2 was added to a culture of *E.coli*. Sample of the culture were then taken out at intervals to determine the number of free T2 virus present.

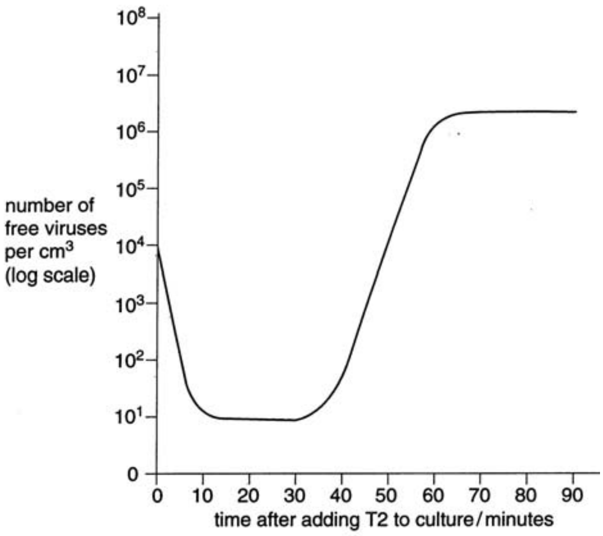

Fig. 3.2

With reference to Fig. 3.2, explain the changes in number of free T2 viruses

(i) In the first 10 mins;

- **No. of free viruses decreases from 10^4 to 10^1 per cm^3;**
- **Due to attachment of viruses on the *E. coli*;** [2]

(ii) between 30 and 60 minutes.

- **No. of free viruses increases from 10^1 to 10^6 per cm^3 ;**
- **Multiplication of viruses;**
- **Lysis of host cell to release viruses;** [3]

(e) Suggest with reasons, a possible evolutionary origin of viruses.

- **Viruses represent escaped nucleic acid from living organisms;**
- **Host cell provided virus with protein coat to isolate them from host cytoplasm;**
- **Virus then evolved the ability to reproduce and penetrate other cells;**
- **Capable of replicating independently from which it originates;**
- **by means of parasitism;** [any 2]

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QUESTION 4

When a red flowered, three-leaved clover plant was crossed with a white flowered, five-leaved plant the following offspring were produced:

red-flowered, five leaved	36
red-flowered, three-leaved	41
white-flowered, five-leaved	40
white-flowered, three-leaved	38

The alleles for red flowers and five leaves are dominant.

(a)

(i) Using the symbols **R** for red flowers, **r** for white flowers, **L** for five leaves and **l** for three leaves, draw a genetic diagram to explain these results. [5]

Parental phenotype	Red flowered, three-leaved clover plant		White flowered, five-leaved plant	
Genotype	**Rrll**		**rrLl;**	
	meiosis			
Gametes	**Rl** **rl**		**rL** **rl** ;	
	meiosis			
Gametes	**Rl**		**rl**	
rL	**RrLl** Red flowered, five -leaved		**rrLl** White flowered, five -leaved	
rl	**Rrll** Red flowered, three -leaved		**rrll;** White flowered, three -leaved	
F1 genotype	**1 RrLl**	**1 Rrll**	**1 Rrll**	**1 rrLl ;**
F1 phenotype	1 Red flowered, five -leaved	1 Red flowered, three -leaved	1 White flowered, three -leaved	1 White flowered, five -leaved ;

(ii) When these plants were self-pollinated, explain why only the white-flowered, three- leaved plants bred true.

- **white-flowered, three-leaved plants were double homozygous recessive;**
- **All the other plants would have been heterozygotes;** [2]

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(b) Use the chi-squared (χ^2) test, and the table of probabilities shown in Table 1, to find the probability of the results of the cross in (a) departing significantly by chance from expectation. Show your working. [2]

χ^2 test $\qquad \chi^2 = \sum \frac{(O-E)^2}{E} \qquad \nu = c - 1$

Key to symbols

$\sum$ = 'sum of …' $\qquad$ O = observed 'value'
ν = degrees of freedom $\qquad$ E = expected 'value'
c = number of classes

Distribution of χ^2

Degrees of freedom	***Probability, p*** 0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.60	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.34	16.27
4	7.78	9.49	11.67	13.28	18.46
5	9.24	11.07	13.39	15.09	20.52

Table 4.1

Penotype	**Observed (O)**	**Expected (E)**	**$(O-E)^2$**	**$(O-E)^2/E$;**
Red flowered, five -leaved	**36**	**38.75**	**7.5625**	**0.195**
Red flowered, three -leaved	**41**	**38.75**	**5.0625**	**0.131**
White flowered, five -leaved	**40**	**38.75**	**1.5625**	**0.040**
White flowered, three -leaved	**38**	**38.75**	**0.5625**	**0.015**
				χ^2 = 0.381

degrees of freedom = n - 1
= 4 – 1 = 3; [2]

(i) Calculated χ^2 = **0.381** $\qquad$ Probability = **> 0.05;** [1]

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(ii) State the conclusion that can be drawn from your calculation of χ^2.

- **Accept null hypothesis.**
- **Observed results are not significantly different from expected results;** [1]

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QUESTION 5

One way of investigating relationships between different primate groups is to study differences in the amino acid sequences of proteins such as haemoglobin.

(a) Explain the origin of differences in the amino acid sequences in a protein.

- **caused by point mutation;**
- **changes the sequence of DNA bases / changes nucleotide / codon;**
- **new triplet codes for a different amino acid;**
- **insertion/deletion causes frameshift;**
- **codes for new polypeptide;** [2]

The β chain of haemoglobin has been investigated and shown to have only four points of difference in all primate groups. The amino acids found at these positions in certain primates or primate groups are shown in the table below.

Primate or primate group	**Amino acid at position 80**	**Amino acid at position 87**	**Amino acid at position 109**	**Amino acid at position 125**
Chimpanzees	Aspartic acid	Threonine	Arginine	Proline
Gibbons	Asparagines	Lysine	Arginine	Glutamine
Old World monkeys	Aspartic acid	Glutamine	Lysine	Glutamine
New World monkeys	Aspartic acid	Glutamine	Arginine	Glutamine
Lemurs	Aspartic acid	Glutamine	Threonine	Alanine

Table 5.1

(b) (i) With reference to Table 5.1, state the two groups that seem to be closely related and give a reason for your answer.

- **Old World monkeys and New World monkeys;**
- **Only one difference / three common amino acids;** [2]

[1 mark each]

(ii) Gibbons are usually classified with the chimpanzees as apes, in the Hominoidea. Explain why the results shown do not support this classification.

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- **Gibbon more like New World monkey;**
- **Gibbon β chain has only one common amino acid (arginine) with chimpanzee;**
- **2 common amino acids with New World monkeys;**
- **Chimpanzee has only common amino acid with / 3 different from each of the other groups;** [3]

[Any 3 - 1 mark each]

(c) Describe four ways, other than differences in amino acid sequences, in which evolutionary relationships between hominids can be investigated. [4]

- **ref to homology / skeletal / skull / dentition details;**
- **carbon dating of fossils allows us to determine the age of a formerly living thing;**
- **anatomical comparisons of fossil primates;**
- **anatomical comparisons of the skeletons of modern species;**
- **sequencing / profiling DNA / DNA hybridization / ref. to PCR / DNA fingerprinting**
- **antibody-antigen / immunological / serum studies;**
- **use of antibodies / agglutination/ immuno-precipitation;**
- **comparison of early stages of animal development reveals additional anatomical homologies not visible in adult organisms;**

[Any 4 - 1 mark each]

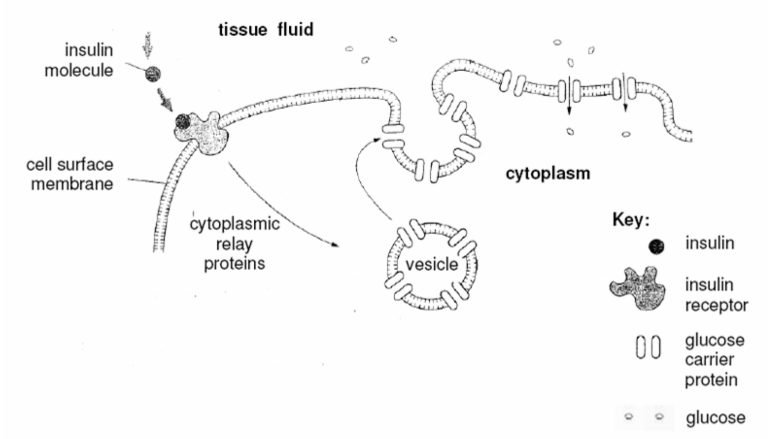
[Total: 11]

QUESTION 6

Type 2 diabetes is a disease condition that is usually caused by the failure of target cells to respond to insulin. This is due to a defect in the insulin response mechanism of the target cells.

Fig. 6.1 shows the normal sequence of cell signaling events when insulin reaches its target cell and binds to an insulin receptor.

Fig. 6.1



(a) State the source of the insulin molecule in Fig. 6.1.

- **Beta cells of the pancreatic islets of Langerhans;** [1]

  (**Reject**: β-pancreatic cells / β-islets)

(b) The insulin receptor is a protein. Suggest how this may be a genetic risk factor for Type 2 diabetes.

- **Harmful mutation of the gene encoding the insulin receptor (reject: insulin) may occur;** [1]

  (Accept: reference to harmful effects of the mutation eg. alteration of specific 3-D conformation of insulin receptor or defective / non-functional / malfunctional receptor or absence of protein receptor altogether;

  Reject: genetic defect / defective gene / faulty gene ie word "mutation" not mentioned.) mutant allele / gene is **transmitted to / inherited by future generations;**

Figure 6.2 shows the structure of the insulin receptor.

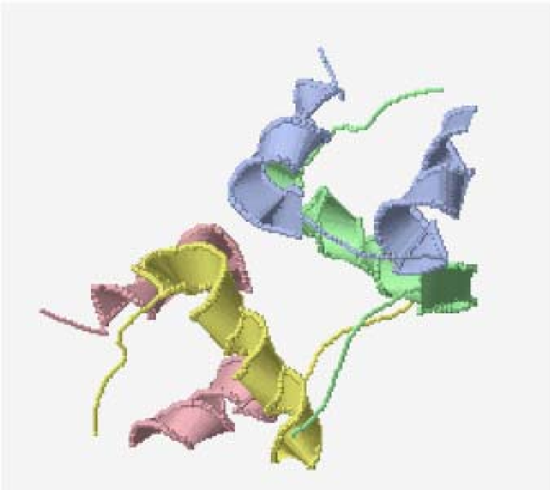


Figure 6.2

(c) With reference to Fig. 6.1 and 6.2,

(i) explain how the binding of insulin to its receptor results in activation of tyrosine kinase.

- **Binding of insulin causes the two receptor subunits / polypeptide chains / monomers / molecules (reject: two receptors) to form a dimer / dimerise;**
- **dimerisation activates the tyrosine kinase region of each subunit;**
- **leading to (auto)phosphorylation of tyrosine residues on the tail of subunits;** [3]

[1 mark each]

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(ii) describe the events after the phosphorylation that lead to an increase in the number of glucose channels in the cell surface membrane.

- Phosphorylation cascade / downstream proteins activated / signal transduction / final activated relay proteins);
- that is amplified;
- causing vesicles carrying glucose carrier proteins / transporters (reject: glucose channels);
- to move / migrate (reject: diffuse) towards cell surface membrane;
- and fuse (accept: mention of exocytosis; reject: bind) with it;
- carrier proteins are then incorporated / inserted into or embedded in membrane (reject: released / added to membrane) as glucose channels; [2]

[Any 4 - 1/2 mark each]

(d) Type 2 diabetes patients are not able to regulate their blood glucose levels owing to the loss of their cells' sensitivity to insulin.

(i) If an increase in blood glucose concentration is not corrected for some time, symptoms of diabetes such as excessive thirst may occur. Suggest how excess blood glucose may cause excessive thirst.

- Glucose causes water potential of the blood to become more negative (reject: lower / decrease); OR
- so that water moves out of cells leading to excessive thirst; [1]

(ii) Explain how the blood insulin level is homeostatically regulated in a normal individual.

- Hyperglycaemia / increased blood glucose concentration stimulates the secretion / release of insulin, the level of which rises;
- Insulin accelerates / promotes / enhances glucose uptake / absorption in target cells;
- The consequent decrease in blood glucose concentration back to normal levels;
- exerts negative feedback; (note: negative feedback must be mentioned in the context of controlling blood insulin, not glucose) on the pancreatic ß cells to inhibit insulin secretion / release;
- cells activated to secrete glucagon which inhibits insulin secretion / release; blood insulin level then falls; [4]

[Any 4 – 1 mark each]

Note: If reference made to "production / synthesis" of insulin / glucagon throughout answer, penalise once.

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QUESTION 7

Fig. 7.1 is an electron micrograph which shows the axon of a motor neuron in transverse section.

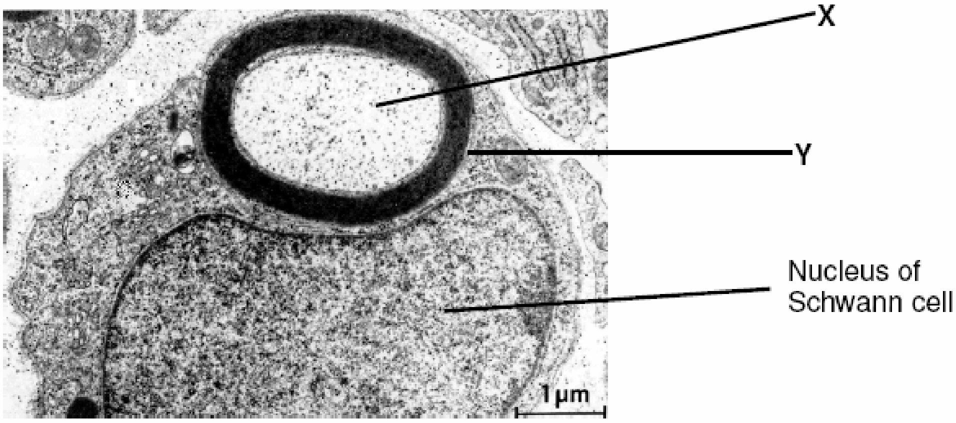

Fig. 7.1

(a) Label the structures X and Y.

X: axoplasm;
Y: myelin sheath/ nurilemma; [1]
[1/2 mark each]

(b) Explain the significance of Y in electrical transmission of nervous impulses.

- **Y acts as an electrical insulator, preventing the loss of current from the axon during electrical transmission;**
- **In a myelinated neurone, action potentials are regenerated only at the nodes of Ranvier / propagated by saltatory conduction;**
- **thus increasing the speed of electrical transmission;** [3]

[1 mark each]

Often present at the synaptic knob of many motor neurons is a large number of mitochondria.

(c)(i) State two uses for the ATP produced by these mitochondria in the synaptic knob of a motor neurone.

- **ATP is required for the Na^+/K^+ pump to work. Na^+ ions are pumped out while K^+ ions are taken into the neurone via active transport;**
- **ATP is required for the action of Ca^{2+} pumps which actively transport Ca^{2+} ions out of the neurone;**
- **ATP is used for the transport / movement of synaptic vesicles towards the membrane of the synaptic knob for the release of neurotransmitter;**
- **ATP used for the absorption / endocytosis of choline and acetate back into the synaptic knob;**

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- **ATP is used to synthesize acetylcholine from choline and acetate;** [2]

[Any 2 – 1 mark each]

(c)(ii) Suggest why during a period of intense nervous activity, the metabolic rate of a nerve cell increases.

- **During intense activity, more action potentials are generated;**
- **Increased activities of Na^+ - K^+ pumps needing more ATP;** [2]

[1 mark each]

(d) Predict the effect on an action potential when the external concentration of sodium ions is lowered.

- **Less sodium ions enter the neurone;**
- **Hence lower action potential;** [2]

[1 mark each]

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Section **B:** Free Response Questions
Answer only **one** question.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections (a), (b) etc., as indicated in the question.

8. **(a)** Outline the role of NAD and NADP in cells. [7]

NADP/ NAD as
- Electron/hydrogen carrier or coenzymes;
- Carry hydrogen atoms from one compound to another;
[max1, ½ m each]

NADP in photosynthesis
- Final electron acceptor in non-cyclic photophosphorylation;
- Drives photolysis of water;
- $2H^+ + 2e^- + NADP \rightarrow NADPH$ or reduced NADP/ hydrogen ions from splitting of water combine with NADP to form NADPH;
- NADPH passes to Calvin cycle/ light independent reactions;
- in the stroma;
- NADPH reduces glycerate bisphosphate/1,3 diphosphoglycerate to triose phosphate/ glyceraldehyde phosphate and gets reoxidized back to NADP;
[max2, ½ m each]

NAD in respiration

Aerobic respiration:
- **Glycolysis: Accept H^+ from triose phosphate/ glyceraldehyde phosphate, which is oxidized to 1,3 diphosphoglycerate or pyruvate;**
- **Forms 2 NADH for each glucose molecule;**
- **Link reaction: Accept H^+ from pyruvate, which is converted to acetyl CoA;**
- **Forms 2 NADH for each glucose molecule;**
- **Krebs cycle: Accept H^+ from intermediates in the regeneration of oxaloacetate from citrate;**
- **Forms 6 NADH for each glucose molecule;**
- **Link rxn and Krebs cycle occurs in the mitochondrial matrix;**
- **Carry hydrogen/ electrons to ETC;**
- **Oxidative phosphorylation/ transfer of electrons down the ETC to release ATP;**
- **1 NADH yields 3 ATP;**
- **NAD reoxidised;**

Anaerobic respiration:
- **NAD regenerated when electrons from NADH transferred to pyruvate/ ethanal;**
- **NAD reused in glycolysis to generate 2 ATP;**

[max4, ½ m each]

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Certain microbes cause disease by disrupting G-protein signaling pathways.¶
The cholera bacterium, *Vibrio cholerae*, present in water contaminated with human feces colonizes the small intestine and produces a toxin that modifies a G protein.¶
¶
The toxin results in a high concentration of cAMP which causes the intestinal cells to secrete large amounts of water and salts into the intestines, leading to profuse diarrhea and death from loss of water and salts.¶
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(b) Describe how photophosphorylation differs from oxidative phosphorylation. [7]

Feature	photophosphorylation	oxidative phosphorylation
Organelle	Chloroplast	Mitochondria;
Location	Thylakoid membrane	Inner Mitochondrial membrane;
Proton reservoirs	Thylakoid space	Intermembrane space;
Energy conversion	Light to chemical	Chemical to chemical;
Role of coenzymes	NADP is reduced in the light rxn as final electron acceptor	NAD is reduced in glycolysis, link rxn and Kreb's cycle;
Sources of electrons	Water	NADH and FADH;
Roles of O_2	By-product of photolysis	Final electron acceptor;
Roles of water	Photolysis: To produce H^+ which will then accumulate in the thylakoid space and electrons to replace the electron lost at the primary reaction centre	By-product of O_2 as the final electron acceptor;

[Any 7 – 1 mark each]
Statements must be comparative.

(c) Describe the roles of membranes in photosynthesis and in respiration. [6]

Photosynthesis
Photosystems / antennae complex / chlorophyll in thlakoid membranes;
ETC, stalked particles (ATP synthase) in thylakoid membranes;
Thylakoid membranes are ion-impermeable / ref. to H^+ gradient;
Ref. to thylakoid membrane separating compartments (compartmentalization);
Ref. to double membrane / envelope;
Ref. to enzymes and Calvin cycle in the stroma; [max 3]

Respiration
Folding of cristae to increase surface area so that there are more stalked particles;
More ETC for oxidative phosphorylation;
ETC and stalked particles in inner membrane / cristae;
Inner membrane ion impermeable / ref. to proton gradient;
Ref. to intermembrane space acting as reservoir of H^+;
Ref. to outer membrane as a barrier;
Ref. to enzymes and Krebs cycle in matrix; [max 3]

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9. (a) Discuss why the structural features of the eukaryotic genome are considered to be more complex than those of the prokaryotic genome. [8]

Eukaryotic	Prokaryotic
Larger number of chromosomes and bases - larger number of genes / coding regions;	**Smaller number of chromosomes and bases** - smaller number of genes / coding regions;
Multiple chromosomes - usually in diploid; - pairing of homologous chromosomes; allows for genetic recombination; during the meiotic form of cell division, - gives rise to genetic diversity;	**Single chromosome** - usually haploid (ie only 1 set of chromosomes); - no pairing of homologous chromosomes possible; hence no genetic recombination as a means of giving rise to genetic diversity;
Presence of non-coding regions - presence of repetitive DNA, transposons etc; - function as yet unknown, may play a role in gene regulation;	**Absence / very little non-coding regions** -most of the genome is coding regions, hence little regulatory elements possible to be present;
Monocistronic genes - individual enzymes are controlled individually by separate promoters; - can be produced in different amounts or not at all;	**Polycistronic genes / operons** - allows coordinated control of metabolic enzymes involved in the same pathway; - all products of operons produced in the same amount;
Presence of control elements - fine tuned control of gene expression, allows for different "levels" of expression of a particular gene depending on the enhancers / silencers; present in the cell then, as well as the strength of the gene's promoter; - response to extracellular environment (Accept: transcription factors, cell signaling);	**Absence of control elements** - allows only crude, single-step control of gene expression; – promoter strength and operator state;
Presence of introns - allows alternative splicing; - presence of introns allows for choice of splice sites, giving rise to multiple mature mRNA molecules and hence multiple possible protein variants from the same gene;	**Absence of introns** - no alternative splicing possible; - hence only one possible protein product per gene;
Association with histones / scaffolding Proteins - allows for increased structural complexity/folding to higher degree of condensation; - control of rate of gene transcription by allowing for increased DNA condensation/conversion between euchromatin and heterochromatin states;	**No association with histones** - does not achieve same level of condensation complexity as eukaryote;

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Deleted: Discuss why the structural features of the eukaryotic genome are considered to be more complex than those of the prokaryotic genome.

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Gene amplification is possible **- increased copy number of genes allows for increased transcriptional levels;**	**Gene amplification does not occur** **- no mechanism for increasing copy numbers of genes;**
Enclosed by nuclear membrane **- no simultaneous transcription and translation; allowing for more levels of control of gene expression;**	**No nuclear membrane** **- simultaneous transcription and translation; hence fewer levels of control of gene expression;**

No marks awarded if features of genomes are mentioned without discussion of their complexity.
No marks awarded if no direct comparison is made.

(b) Compare and contrast the control of gene expression in prokaryotic and eukaryotic cells. [6]

Differences (max = 3)

Eukaryotic	Prokaryotic
Controls at chromosomal level **- gene amplification occurs;** **- histone modifications can occur, resulting in conversion between euchromatin and heterochromatin, and hence the ease of transcription;**	**No controls at chromosomal level** **- no gene amplification occurs;** **- DNA not associated with histones, hence no histone modification possible;**
Controls at transcriptional level **- Three different classes of RNA each synthesized by a different RNA polymerase;** **- control elements eg enhancer and silencer also involved in controlling rate of transcriptional initiation; control elements may be proximal or distal**	**Controls at transcriptional level** **- All RNAs synthesized by the same RNA polymerase;** **- Operator involved in control of transcription; operator is proximal to the genes under its control** **- inducible and repressible operons; some operons undergo both positive and negative regulation**
Monocistronic mRNAs **- Each gene controlled by its own promoter;**	**Polycistronic mRNAs** **- Multiple related genes controlled by the same promoter (operon system);**
Controls at post-transcriptional level **- Nuclear membrane separates the processes of transcription and translation;** **- post transcriptional modifications (addition of 5'cap, 3' poly-A tail, splicing) occur;** **- 5'cap and 3' poly-A tail increase transcript stability / prevent transcript degradation;** **- alternative splicing possible;**	**No controls at post-transcriptional level** **-Transcription and translation occur simultaneously; due to the absence of a nuclear membrane;** **- no post transcriptional modifications occur;** **- leads to lower stability of transcript and its degradation; within seconds / minutes**
Controls at translational level **- control of rate of translational initiation;**	**No control at translational level** **- control at this level is unlikely; due to simultaneous transcription and translation**

Controls at post-translational level	No controls at post-transcriptional level
- post-translational modifications (eg glycolsylation, cleavage, phosphorylation etc); determine the functional abilities of the protein	- no/minimal post-translational modifications occur;

Similarities (max = 3)

- Most significant / main level of control at transcriptional level;
- Rate of transcriptional control is affected by the strength of the promoter;
- Rate of transcriptional initiation can be affected by protein binding to regulatory elements;
  (repressors to operators in prokaryotes, activators and repressors to enhancers and silencers respectively in eukaryotes – ie ability to respond to external signals)
- Universal genetic code in prokaryotes and eukaryotes;
- Polyribosomes speed up the rate of translation;
- Degradation of mRNA / proteins as a method of terminating response;

(c) Describe the structure and function of the portions of eukaryotic DNA that do not encode for protein or RNA. [6]

- In eukaryotes, about 97% of DNA does not code for proteins or RNA;
- They consist of regulatory sequences, introns and repetitive DNA;

UTR

- UTRs are untranslated regions found at the 5' and 3' ends of most mRNAs;
- 5' UTR is also known as a leader sequence;
- contains a ribosome binding site;
- contains binding sites for proteins that affects mRNA's stability;
- prokaryotic mRNA's 5' UTR is usually short;
- some viruses have long 5' UTR which may be important in gene expression;

Promoter

- DNA sequences found at the 5' end adjacent to transcriptional site;
- immediately upstream of coding region of gene;
- RNA polymerase and transcription factors bind to the promoter;
- To initiate production of mRNA transcript;
- Interactions of proteins at the promoter regulate gene activity;
- By activating or repressing transcription;
- An eg. of a 6 promoter seq. is the "TATA box"/ TATAAA;
- Found about 25-35 nucleotides before the transcriptional site;

Control elements

- Are noncoding DNA segments that regulate transcription by binding transcription factors;
- Proximal control elements are located close to the promoter;
- Distal control elements or enhancers may be far away from a gene;
- enhancer / silencer sequences have base sequences that can be recognized by activator / repressor proteins respectively;

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- interaction with promoter occurs via protein-mediated DNA bending;

Repetitive DNA

- about 10% of mammals' genome is tandemly repetitive DNA/ satellite DNA;
- much of the satellite DNA play a structural role at telomeres and centromeres;
- Telomeres - Protects the ends of chromosomes from hydrolysis by cellular exonucleases;
- Centromeres - region where sister chromatids join in a replicated chromosomal structure during cell division;
- anchor point for spindle formation required for the pull of chromosomes toward the centrioles during anaphase;
- allows for equal separation of genetic material to daughter cells during anaphase of cell division;
- absence of centromere results in nondisjunction of chromosomes; aneuploidy;

Transposons

- Genes that move from one location to another within the genome;
- If it "jumps" into a coding sequence, it can prevent normal gene function;

Telomeres

- Found at ends of linear chromosomes;
- Long stretches of tandemly repeated;
- that fold over to form hairpin loops;
- Protects the ends of chromosomes from hydrolysis by cellular exonucleases;
- Prevents spontaneous fusion of the ends of linear chromosomes;
- Protects coding regions of DNA from loss during semi-conservative replication (end replication problem);
- Serves as signals for cells to enter replicative senescence;
- preventing excessive cell divisions after a critically short length is reached;
- absence of telomere control results in uncontrolled cell proliferation;

Centromeres

- constricted region which is transcriptionally inactive;
- short, tandemly repeated;
- DNA sequences known as alpha satellite DNA;
- region where sister chromatids join in a replicated chromosomal structure during cell division
- kinetochore formation takes place;
- anchor point for spindle formation required for the pull of chromosomes toward the centrioles during anaphase;
- allows for equal separation of genetic material to daughter cells during anaphase of cell division;
- absence of centromere results in nondisjunction of chromosomes; aneuploidy;

Students should have a range of points.
[1 mark each]

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Nov 2003 Paper 2 Q1

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Collagen is found in the extracellular matrix of muscles, tendons, ligaments and bones. Fibroblast cells in these tissues make collagen by synthesizing polypeptides that form molecules with a triple helix shape. These are secreted from fibroblasts into the extracellular matrix where enzymes assemble them into collagen fibres.

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VJC Promo 2006 Q1d

(a) State two similarities between rough endoplasmid reticulum and golgi apparatus.

[2]

(b) State two differences between rough endoplasmid reticulum and golgi apparatus.

[2]

Similarity *(any one)*

1. Both consist of **membrane**-bound **sacs/tubules** called cisternae. [1]
2. Both are made up of **single membrane**; [1]
3. Both produce vesicles that bud off the organelle; [1/2]

Differences

Structure D (rER)	Structure F (GA)
1. Consist of a <u>network</u> of <u>interconnected</u> membrane-bound sacs and tubules. [1/2] NB: idea of interconnectedness is impt.	Consist of <u>stacks</u> of <u>flattened</u> membrane-bound sacs. [1/2]
2. Presence of ribosomes. [1/2] ® ribosomes *may or may not* be present	Absence of ribosomes. [1/2]
3. Connected to the outer **membrane** of the nucleus. [1/2]	3. Not connected to the outer **membrane** of the nucleus. [1/2]
4. Does not have a definite forming and maturing face.	4. Has a definite forming face where the vesicles join and a maturing face where the vesicles bud off.

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Fig. 4.1 is a diagram summarizing these events.

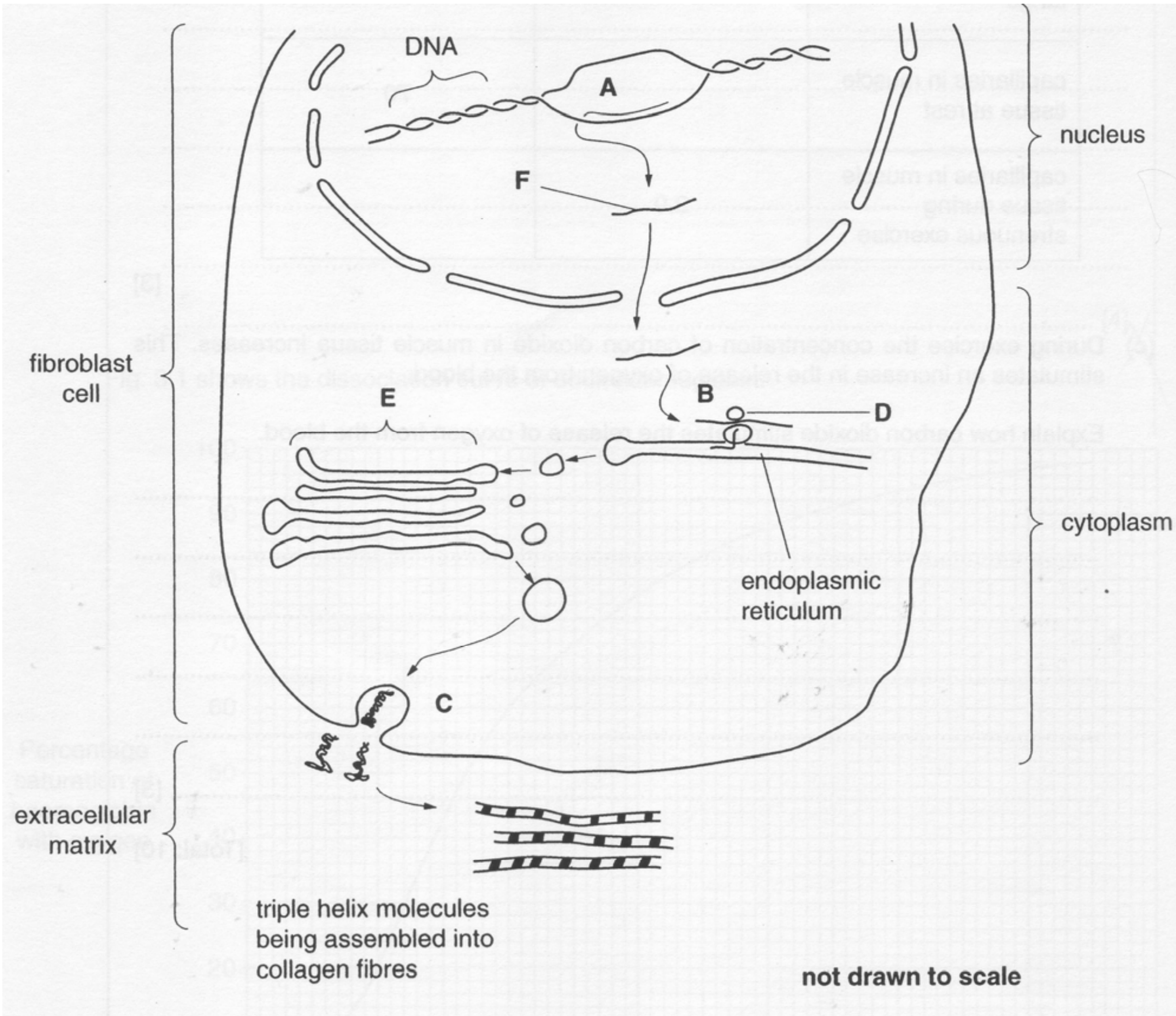

Fig. 4.1

(a) (i) Name the processes occurring at **A**, **B** and **C**. [3]

A Transcription
B Translation
C exocytosis

(ii) Name structures **D** and **E**. [2]

D Sub unit of ribosome
E Golgi apparatus/body

(iii) Name molecule **F.** [1]

F mRNA

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a- to remove diagram and combine with something else….

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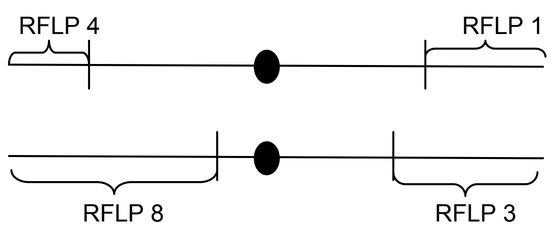
Huntington's Disease (HD) is a neurodegenerative disorder due to an autosomal dominant allele. The disease is characterized by progressive motor dysfunction and intellectual deterioration but its symptoms usually occur when individuals are in their 30's-50. To circumvent the problem, RFLP has been used these days for early detection of the disease.

Patients with HD are known to have RFLP 1 and the diagram below shows a pair of homologous chromosomes from a male.



An offspring of the male has a chromosome with RFLPs 1 and 8, contributed by the father.

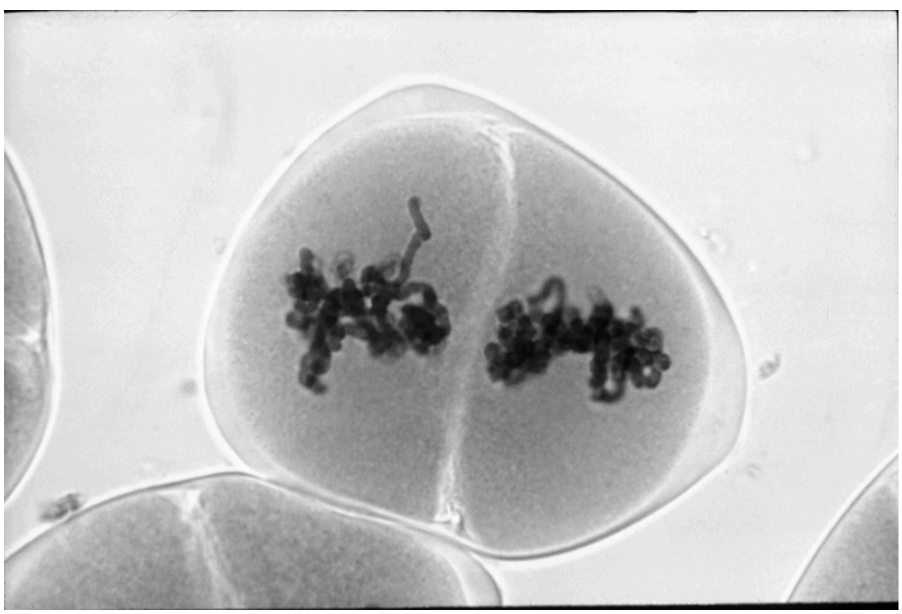
(a)(i) State the event that must occur during meiosis in order to obtain a chromosome
with RFLPs 1 and 8.
[1]

____

(ii) In the space below, draw the event occurring in (a).
[3]

(iii) Suggest a reason why Huntington's Disease is likely to persist in populations. [1]

The micrograph below shows a Lily cell undergoing cell division.



[© Clare Hasenkampf/BPS]

(b) Refer to the micrograph above and:

(i) Identify the type and stage of nuclear division.

[2]

(ii) Briefly explain the reason for your answer in (b)(i).
[2]

____

____

____

____

(c) Describe the roles of spindle fibers in meiosis.
[2]

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between non-sister chromatids

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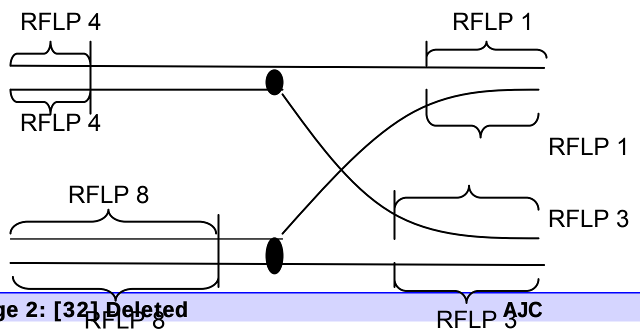

[1] - 2 sister chromatids per chromosome
[1] – a crossing over of non-sister-chromatids; may also take place on the other arm betw RFLP 4 and 8labelling of centromere (1/2) and chromatids/chromosomes (1/2)

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Metaphase bec:
metaphase bec chromosomes are aligned at equator

metaphase 2 bec:
2 metaphase plate has formed → meiosis
OR

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the daughter cell have chromosomes aligned at the new equatorial plate which is at right angles to the first

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Other (non-kinetochore) spindle fibres lengthen to elongate the cell in preparation for cytokesis

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X:
Y:

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Describe another mechanism which regulates the expression of β-galactosidase at Y.
[2]

(d)

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(i) S

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(ii) The bacterial cells were found to be viable. Suggest how the bacteria can survive without a carbon source. [1]

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(a) State the two mechanisms of regulation of lac operon. [2]

negative control by the lac repressor
positive control by cAMP receptor protein.

(b) With reference to Fig.1, explain how expression levels of _ -galactosidase is regulated by
the **lac repressor** from time X to Y. [6]

X:

Both glucose and lactose are present and glucose is used but not lactose
In the presence of lactose, allolactose binds to lac repressor. Thus, the Lac repressor
is in its inactive configuration and does not bind to the operator
But another mechanism prevents transcription of the lac operon due to catabolite
repression (ie. CRP is inactive and disengages from CRP binding site)
RNA polymerase is unable to bind to the promoter/Low _ –galactosidase mRNA is produced

Y:

Once the glucose concentration has depleted, *E. coli* uses Lactose.
Allolactose binds to the lac repressor
Lac repressor is inactive and does not bind to the operator

RNA polymerase is able to bind to the promoter to transcribe the mRNA of the lac
operon / Lac operon is switched on / High _-galactosidase mRNA is produced

(c) Describe another mechanism which regulates the expression of _-galactosidase at Y. [2]

In low levels of glucose, cAMP levels increases , CRP is active and it binds to the CRP binding site.
Transcription of B-gal proceeds at a high level.

(d)(i) Suggest one mechanism that causes the amount of _-galactosidase to decrease. [1]

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mRNA degradation/ protein is degraded
(ii) The bacterial cells were found to be viable. Suggest how the bacteria can survive without
a carbon source. [1]

Stress genes can be turned on which allows the bacterial cells to survive. Under
stress or starvation (starvation sigma factor)

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In an experiment to investigate the function(s) of capping and polyadenylation, scientists introduced firefly luciferase mRNA, with and without a cap, and with and without poly (A), into tobacco cells. The amount of remaining luciferase mRNA and the luciferase activity were measured experimentally. Luciferase is a protein product that is easy to detect because of the light it generates in the presence of appropriate substances (luciferin and ATP).

The table below illustrates the effects of capping and polyadenylation on the lifespan of the luciferase mRNA and its protein products.

Table 3.1

mRNA	**Luciferase mRNA Half Life (min)**	**Luciferase Activity (units)**
Uncapped		
Poly(A) tail absent	31	2,941
Poly(A) tail present	44	4,480
Capped		
Poly(A) tail absent	53	62,595
Poly(A) tail present	100	1,331,917

Note : Half life refers to the time taken for the amount of mRNA to be degraded to half the original amount

(d) (i) Explain why luciferase activity can be used as a reflection of translation efficiency. [2]

With reference to the Table 3.1, describe the effect of polyadenylation on mRNA degradation in capped mRNA. [2]

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With reference to Table 3.1, describe the effects of capping and polyadenylation on translation efficiency. [2]

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(d) (i) Luciferase (protein) is the end product of translation;
easily detected as a light / measurable [2]

(ii) Increase/ doubles mRNA half-life;
from 53 units to 100 units; [2]

(iii) Capping increases translation efficiency between 21 to 297 folds;
Polyadenylation increases translation efficiency between 1.5 to 21 folds [2]

(e) 5 **S**mall **n**uclear **R**ibonuclear **P**roteins (snRNPs) associate with other proteins; form a **spliceosome**;
RNA-protein interactions create a **lariat** structure;
Spliceosome recognizes the splicing signals/ splice site/ short nucleotide sequences at the end of intron on a pre-mRNA molecule;
Lariat structure excises introns;
Joins exons together to form mature mRNA;

c and d ii too specific but can use 1st part.

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Genetic Variation (Crossing)
Nov 2006 Paper 2 Q4

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A cross was made between two fruit flies. One parent had an unknown genotype since it had the wild phenotype (nomal [long] wings and normal body colour). The other parent had vestigial [undeveloped] wings and ebony body colour, so it was known to be homozygous recessive for both characteristics.

The resulting offspring were as follows:

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normal wings and normal body colour

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normal wings and ebony body colour

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vestigial wings and normal body colour

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vestigial wings and ebony body colour

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Using the following symbols:

N normal wings **n** vestigial wings
E normal body **e** ebony body

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Draw a genetic diagram in the space below to show the cross described.

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vestigial [undeveloped] wings and ebony body colour

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nomal [long] wings and normal body colour

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E = expected ‘value’

ne;

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normal wings and normal body colour

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normal wings and ebony body colour
vestigial wings and normal body colour

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vestigial wings and ebony body colour;

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Explain why it would be useful to carry out a chi-squared test on these results. No calculations are required to answer this question.

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[2]

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Expected results is a ratio of 1:1:1:1;
Whether observed results differed significantly from expected results with a less than 5%;

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Vestigial wing and ebony body are caused by artificially induced mutations in laboratory bred fruit flies. Suggest why such mutants are unlikely to be found in natural populations.

[2]

Vestigial wing and ebony bodied flies would be disadvantaged in the wild because they cannot fly from and are more visible to predators and therefore unlikely to survive as they are more likely to be eaten;
Mutations are rare and such alleles are not passed on to offspring because they are unlikely to reproduced;
Such characters are coded for by recessive alleles which are therefore masked by dominant alleles;

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Answers

(a) caused by point mutation;
changes the sequence of DNA bases / changes nucleotide / codon;
new triplet codes for a different amino acid;
substitution causes frameshift;
codes for new polypeptide [2]

(b)(i) Old World monkeys and New World monkeys;
Only one difference / three common amino acids [2]

(b)(ii) Gibbon more like New World monkey;
Gibbon β chain has only one common arginine / amino acid with chimpanzee;
2 common amino acids with New World monkeys;
chimpanzee has only common amino acid with / 3 different from each of the other groups [3]

(c) anatomical comparisons of fossil primates;

dating of fossils;
sequencing / profiling DNA / DNA hybridization / ref. to PCR / DNA fingerprinting
anatomical comparisons of the skeletons of modern species;
ref to homology / skeletal / skull / dentition details;
antibody-antigen / immunological / serum studies;
use of antibodies / agglutination/ immuno-precipitation; [any 2]

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modify the peak for fig. 6.2 at 1st part…

6 *What do you understand by the following terms?*

(a) i. *Species [2]*
- Any group of organisms;
- Capable or potentially capable of interbreeding with one another;
- To produce fertile offspring;
- Show any similarities in evolution, ecology, morphology, behavior, biochemistry (any 2);

ii. *Genetic variation [1]*
- Members of a population;
- Differ in one or more inherited characteristics;

(b) *What impact would low genetic variability in a population of this species have on species survival? Explain your answer. [2]*
- Low number of alleles would reduce probability of the presence of a favorable characteristic in the population;
- If there was an environmental change;
- Reduced chances of survival / species extinction;
- Continuous inbreeding between closely related individuals;
- Inbreeding depression may occur;

(c) *Using the information from **Figure 6.1**, explain the variation shown. [2]*
- Polygenes;
- Continuous variation / normal distribution / bell shaped;
- Range seen due to the additive effects of genes / alleles;
- Effects of environment;

(d) *How does the information in **Figure 6.2** support your answer in (c)? [1]*
- Breeding with individuals of similar genotypes;
- Shifted the mean / variation towards either end;

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(e) *What type of selection was shown by **Figure 6.2**? [1]*
- Disruptive selection;

(f) *Suggest what might happen if the high activity score species produced by the breeding experiment are now introduced to the low activity score species. [1]*
- Failure to interbreed; unable to produce fertile offspring;
- Competition between both species; with the survival of stronger species;

(g) *Suggest how aggression probably evolved in this species. [2]*
- Individuals, within population, show variations in behavior, i.e. peaceful and aggressive natures;
- Aggression arises from mutation / a collection of mutations;
- Aggression could have been a beneficial trait to possess, in the particular habitat that the feline species lived in (environmental selection); OR
- Aggressive behavior might be favored by females of the species, causing it to be a selected trait (sexual selection);
- Selected individuals survive longer, reproduce and pass on these favorable traits to their offspring;

(h) *Explain why the feline population is the smallest unit that allows for the evolution of aggression [3]*
- Although Natural selection acts on individuals;
- Affecting its survival and reproductive success compared to other individuals;
- Evolution looks at accumulated changes of populations over time;
- So individuals must be able to interbreed with other members (constituting a population) to produce offspring (next generation);
- Pass on genes to the next generation;
- Aggression (trait) must be inheritable/ Aggression behavior probably controlled by genes;

Alternative answer
- Natural selection acts on individuals;
- Selecting for the feline's aggression traits
- That affects its survival and reproductive success compared to other individuals;
- Aggression behavior probably controlled by genes;
- Long term effects of natural selection are at the level of the gene and the population rather than the individual;
- As it is the members of the population which can interbreed, exchange genes and thereby pass on genes to the next generation;
- Evolution measured as changes in relative proportions of heritable variations in a population in successive generations;

Clement will give an Evolution Q.

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Hormonal Control
HCI Prelim 2007

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The structure of the insulin receptor is shown in Figure 6.2.

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explain how the binding of insulin to its receptor results in an increase in the phosphorylation of serine and threonine residue side-chains of the cytoplasmic relay proteins.

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of serine and
threonine residues of cytoplasmic relay proteins

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Answer:
(a)

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Identify the source of the insulin molecule in **Figure 6.1**. [1]

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cells of the pancreatic islets of Langerhans;;
(**Reject**: b-pancreatic cells / b-islets)

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Note: Award 1m if all 3 points above (in bold) present, deduct 0.5m for every point missing / critical spelling mistake.

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(b) The insulin receptor is a protein.

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be a genetic risk factor for Type 2 diabetes.
[1]

Harmful mutation of the **gene encoding** the **insulin receptor** (reject: insulin) may occur**;**

(Accept: reference to harmful effects of the mutation eg. alteration of specific 3-D conformation of insulin receptor or defective / non-functional / malfunctional receptor;

Reject: genetic defect / defective gene / faulty gene ie word "mutation" not mentioned.) mutant allele / gene is **transmitted to / inherited by future generations;**

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With reference to **Figures 6.1** and **6.2**,
(i)

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explain how the binding of insulin to its receptor results in activation of tyrosine kinase

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in an increase in the phosphorylation of serine and threonine residue side-chains of cytoplasmic relay proteins

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Binding of insulin causes the **two receptor subunits / polypeptide chains / monomers / molecules** (reject: two receptors) to **form a dimer / dimerise;**
dimerisation **activates** the **tyrosine kinase** region of each subunit**;**

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leading to **(auto)phosphorylation of tyrosine residues** on the tail of subunits**;**

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Specific **relay protein(s)** in the cell **bind(s)** to specific **phosphorylated tyrosine(s);** and becomes **activated;**

activated relay protein(s) trigger(s) **transduction pathway(s) / phosphorylation cascade;**

leading to **activation of** serine and threonine **kinases;** / **inhibition of** serine and threonine **protein phosphatases;**

in the cytoplasm. (Accept: serine and threonine kinases / protein phosphatases bind to
phosphorylated tyrosine residues**;** and are activated; / inhibited**;** respectively) (max 1.5m)

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of serine and threonine residues of cytoplasmic relay proteins

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that lead to an increase in the number of glucose channels in the cell surface membrane. [2]

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Phosphorylation of cytoplasmic relay proteins constitutes a transduced **chemical signal (**accept: mention of p

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Nervous Control
TJC Prelim 2007

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Phosphorylation cascade / downstream proteins activated / signal transduction / final activated relay proteins)**;**

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that is **amplified;**
causing vesicles carrying **glucose carrier proteins / transporters** (reject: glucose channels)**;**
to **move / migrate** (reject: diffuse) towards cell surface membrane**;**
and **fuse** (accept: mention of exocytosis; reject: bind) with it**;**
carrier proteins are then **incorporated / inserted into or embedded in membrane** (reject: released / added to membrane) as glucose channels**;**

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causing vesicles carrying **glucose carrier proteins / transporters** (reject: glucose channels)**;**

to **move / migrate** (reject: diffuse) towards cell surface membrane**;**

and **fuse** (accept: mention of exocytosis; reject: bind) with it**;** carrier proteins are then **incorporated / inserted into or embedded in membrane** (reject: released / added to membrane) as glucose channels**;**

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Type 2 diabetes patients are not able to regulate their blood glucose levels owing to the loss of their cells' sensitivity to insulin.

(i)

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If an increase in blood glucose concentration is not corrected for some time, symptoms of diabetes such as excessive thirst may occur. Suggest how excess blood glucose may cause excessive thirst. [1]

Glucose causes **water potential** of the **blood to become more negative** (reject: lower / decrease)**; OR**
so that water moves out of cells

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/ tissue fluid into the blood / loss of water from cells by **(ex)osmosis;**

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leading to excessive thirst.

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(Accept: excessive urination**;** / increase in blood osmolarity**;** stimulates hypothalamus or brain**;**)

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(ii)

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Explain how the blood insulin level is homeostatically regulated in a normal individual.
[3]

Hyperglycaemia / increased blood glucose concentration stimulates the **secretion / release** of **insulin**, the level of which rises**;**

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(note: both primary stimulus and response must be mentioned before 0.5m awarded)

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Insulin **accelerates / promotes / enhances glucose uptake / absorption**

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in target cells;

The consequent **decrease in blood glucose concentration** back to normal levels**;**

exerts **negative feedback;** (note: negative feedback must be mentioned in the context of controlling blood insulin, not glucose, level before 0.5m awarded) on the pancreatic _ cells to **inhibit insulin secretion / release;**

cells activated to secrete **glucagon** which **inhibits insulin secretion / release;** blood insulin level then falls.

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,

increases the rate of glucose storage in liver cells**;**

and **stimulates lipogenesis** in adipocytes**;** (max 1m)

The consequent **decrease in blood glucose concentration** back to normal levels**;**

exerts **negative feedback;** (note: negative feedback must be mentioned in the context of controlling blood insulin, not glucose, level before 0.5m awarded) on the pancreatic _ cells to **inhibit insulin secretion / release;**

_ cells activated to secrete **glucagon** which **inhibits insulin secretion / release;** blood insulin level then falls.

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Note: If reference made to "production / synthesis" of insulin / glucagon throughout answer, penalise once.

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Can students relate metabolic rate to Na-K pumps?

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concentration of sodium ions are

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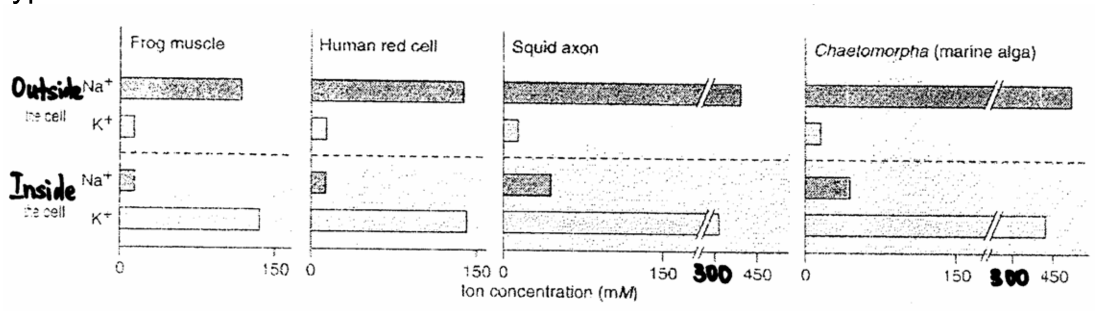
AJC Common Test 2006

(d) Figure 2 below is a series of graphs showing intracellular and extracellular concentrations of sodium (Na^+) and potassium (K^+) ions in several cell types.



The double stroke (//) in the graphs of squid axon and *Chaetomorpha* (marine alga) shows that the graphs are cut.

Figure 2

With respect to Figure 2, state the trends observed in all cell types. [2]

(ii) Which part of the cell allows for the generation of these trends stated in (a) (i) [1]

(iii) Suggest a reason why the maintenance of this distribution of Na^+ and K^+ ions is important to these cell types. [1]

(c) Describe the events that lead to the release of the neurotransmitter from the presynaptic membrane into the synaptic cleft. [2]

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Answers:
Figure 2 is an electron micrograph which shows the axon of a motor neurone in transverse section.

(a)

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Label the structures X and Y. [2]

X: axoplasm Y: myelin sheath

(b)

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Explain the significance of Y in electrical transmission of nervous impulses. [3]
Y acts as an electrical insulator, preventing the loss of current from the axon
during electrical transmission.
In a myelinated neurone, action potentials are regenerated only at the nodes of
Ranvier / propagated by saltatory conduction,
thus increasing the speed of electrical transmission.

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Often present at the synaptic knob of many motor neurons

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are a large number of mitochondria.

(c)(i)

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State two uses for the ATP produced by these mitochondria in the synaptic knob of a motor

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neurone. [2]

Any two:
ATP is required for the Na+/K+ pump to work. Na+ ions are pumped out while K+

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ions are taken into the neurone via active transport.
ATP is required for the action of Ca2+ pumps which actively transport Ca2+ ions out

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of the neurone.
ATP is used for the transport / movement of synaptic vesicles towards the membrane of the synaptic knob for the release of neurotransmitter.
ATP used for the absorption / endocytosis of choline and acetate back into the

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(c)(ii) Suggest why during a period of intense nervous activity, the metabolic rate of a nerve cell increases.
[2]

During intense activity, more action potentials are generated;

Increased activities of Na^+ - K^+ pumps needing more ATP

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Superscript

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(d) Suggest the effect on an action potential when the external concentration of sodium ions are lowered. [2]

Less sodium ions enter the neurone;
Hence lower action potential

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(d) (i)

High Na^+ concentration outside the cell and a high K^+ concentration inside the cell;
Quote any cell's values –

		Ions concentration in			
Location	**Ions**	**Frog muscle**	**Human r.b.c**	**Squid axon**	***Chaetomorpha* (marine alga)**
Outside the cell	Na^+	114	150	400	500
	K^+	14	14	14	14
Inside the cell	Na^+	14	14	42	42
	K^+	135	150	300	400

[Each point → 1 mark]

(ii)

Cell membrane;
Na^+ and K^+ ion channels/ Na^+/K^+ pump;

[Each point → ½ marks]

(iii)

Maintains the resting membrane potential to generate nerve impulse;

OR

Contributes to the electrochemical gradient to establish the resting membrane potential;

[Each point → 1 mark]

(c)

Action potential arrives at the synaptic knob;
Depolarization increases membrane permeability to $Ca^{2+;}$
opening of voltage gated calcium channels and Ca^{2+} influx;
Ca^{2+} influx causes synaptic vesicles to fuse with presynaptic membrane and discharge neurotransmitters by exocytosis;

Underlined words should appear in student's answer.

[Each point → ½ marks]

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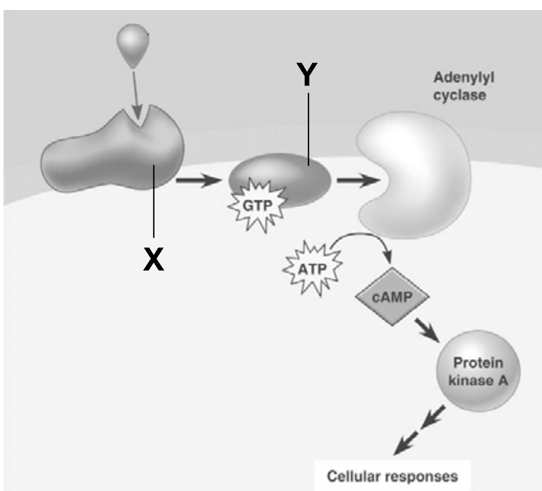
Cell Signaling
SAJC Prelim 2007
QUESTION 8

Certain microbes cause disease by disrupting G-protein signaling pathways. The cholera bacterium, *Vibrio cholerae*, present in water contaminated with human feces colonizes the small intestine and produces a toxin that modifies a G protein.

The toxin results in a high concentration of cAMP which causes the intestinal cells to secrete large amounts of water and salts into the intestines, leading to profuse diarrhea and death from loss of water and salts.



(a) Name molecules **X** and **Y**.

X ……………………..………… **Y** ……………..……………..…..…………………..[1]

(b) What is the general name given to small molecules or ions that function like cAMP in signaling pathways?

……………………………………………………………………………………………..……
………[1]

(c) What is the significance of a phosphorylation cascade in cell signaling?

…………………………………………………………………………..…………………
……..

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……………………………………………………………………………………………..……
………[2]

(d) State the function of adenylyl cyclase.

……………………………………………………………………………………………..……
………[1]

(e) Suggest how the toxin leads to an increase in cAMP concentration.

…………………………………………………………………………..…………………
……..

…………………………………………………………………………..…………………
……..

…………………………………………………………………………..…………………
……..

……………………………………………………………………………………………..……
………[3]

[Total: 8]

(a) Name molecules X and Y. [1]

[½ mark per pt]

X = G-protein linked receptor

Y = G-protein

(b) What is the general name given to small molecules or ions that function like cAMP in signaling pathways? [1]

second messengers

(c) What is the significance of a phosphorylation cascade in cell signaling? [2]

signal amplification
specificity of cell signaling
ref. cross-talk (interaction between different signaling pathways)

(d) State the function of adenylyl cyclase. [1]

convert ATP to cAMP

(e) Suggest how the toxin leads to an increase in cAMP concentration. [3]

modified G protein unable to hydrolyze GTP to GDP
remains stuck in its active form
adenylyl cyclase is <u>continuously</u> stimulated to make cAMP

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Photosynthesis and Respiration
Nov 2006 Paper 2 Q7

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NADP/ NAD as
- Electron/hydrogen carrier or coenzymes;
- Carry hydrogen atoms from one compound to another;
[max1, ½ m each]

<u>NADP in photosynthesis</u>
- Final electron acceptor in non-cyclic photophosphorylation;
- Drives photolysis of water;
- $2H^+ + 2e^- +$ NADP → NADPH or reduced NADP/ hydrogen ions from splitting of water combine with NADP to form NADPH;
- NADPH passes to Calvin cycle/ light independent reactions;
- in the stroma;
- NADPH reduces glycerate bisphosphate to triose phosphate and gets reoxidized back to NADP;
[max2, ½ m each]

<u>NAD in respiration</u>

Aerobic respiration:
Glycolysis: Accept H^+ from triose phosphate, which is oxidized to pyruvate;
Forms 2 NADH for each glucose molecule;
Link reaction: Accept H^+ from pyruvate, which is converted to acetyl

CoA;
Forms 2 NADH for each glucose molecule;
Krebs cycle: Accept H^+ from intermediates in the regeneration of oxaloacetate from citrate;
Forms 6 NADH for each glucose molecule;
Link rxn and Krebs cycle occurs in the mitochondrial matrix
Carry hydrogen/ electrons to ETC;
Oxidative phosphorylation/ transfer of electrons down the ETC to release ATP;
1 NADH yields 3 ATP;
NAD reoxidised;

Anaerobic respiration:
NAD regenerated when electrons from NADH transferred to pyruvate/ ethanal;
NAD reused in glycolysis to generate 2 ATP;
[max5, ½ m each]

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NADPH reduces glycerate bisphosphate to triose phosphate and gets reoxidized back to NADP;

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Coenzyme NAD and NADP are reduced, carry electrons and protons to the electron transfer chain and are regenerated;
NAD is formed in glycolysis, link reaction and Kreb's cycle by oxidation;
in the mitochondrial matrix;
And then used in oxidative phosphorylation;
NADP location is in the stroma of the chloroplast;
NADP is reduced in the light reaction as the final electron acceptor;
And then was used in the Calvin cycle as a reducing agent;

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Nov 2002

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Photosynthesis

Photosystems / antennae complex / chlorophyll in thlakoid membranes;
ETC, stalked particles (ATP synthase) in thylakoid membranes;
Thylakoid membranes are ion-impermeable / ref. to H^+ gradient;
Ref. to thylakoid membrane separating compartments (compartmentalization);
Ref. to double membrane / envelope;
Ref. to enzymes and Calvin cycle in the stroma; [max 3]

Respiration

Folding of cristae to increase surface area so that there are more stalked particles;
More ETC for oxidative phosphorylation;
ETC and stalked particles in inner membrane / cristae;
Inner membrane ion impermeable / ref. to proton gradient;
Ref. to intermembrane space acting as reservoir of H^+;
Ref. to outer membrane as a barrier;
Ref. to enzymes and Krebs cycle in matrix [max 3]

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Explain the effect of carbon dioxide and light as limiting factors on the rate of photosynthesis.

CO_2 is a raw material for the calvin cycle
Its increased concentration increased the rate of photosynthesis
Its level is low (0.04%) in the atmosphere and therefore plays a major role in limiting photosynthesis
Light intensity is also important in the light dependent stage to excite electrons
Chl a absorbs max light at 680nm while chl b absorbs max light at 480nm
Compensation point is the light intensity when rate of phosynthesis and respiration is the same and it is reached at low light intensity
Is therefore rarely limiting during daylight hours
High light intensity also opens the stomata

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Nov 2002 Essay 10c roles of mebms in PS and RS (6)

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Organization and Control of Pro- and Eukaryotic Genome
HCI Prelim 2007

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Anatomical homology

anatomical resemblances representing variations on a structural theme present in their common ancestor

ref. examples:

forelimbs of all mammals, including humans, cats, whales and bats show the same arrangement of bones from the shoulder to the tips of the digits/ vestigial legs in snakes

even though these appendages can have very different functions (lifting, walking, swimming and flying)

Embryology homology

comparison of early stages of animal development reveals additional anatomical homologies not visible in adult organisms

ref. examples:

embryos of different types of vertebrates have a tail posterior to the anus/ pharyngeal (throat) pouches.

these embryonic structures develop into homologous structures with very different functions, such as gill slits in fishes and parts of ears and throat in humans.

Molecular homology

all forms of life use the same genetic machinery of DNA and RNA, and the genetic code is essentially universal.

DNA differences are accumulated as descendents evolve / closely related species share a greater portion of their DNA

ref. examples:

organisms as dissimilar as humans and bacteria share many genes inherited from a distant common ancestor

/ comparison of amino acid sequence of human haemoglobin with other vertebrates reveals the same pattern of evolutionary relationships as nonmolecular methods

In relation to Darwin's theory

ref. similarities between individuals of the same species or among different species, that arose from common ancestry

NOTES:

Darwin's theory = many modern species of organisms are descendents of ancestral species that were different from the species. As the descendents

of that ancestral organism spilled into various habitats over millions of years, they accumulated diverse modifications, or adaptations, that fit them to specific ways of life.

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Describe the neutral theory of molecular evolution.

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(b) (c) Compare and contrast the control of gene expression in prokaryotic and eukaryotic cells.

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disadvantageous mutations are quickly removed by natural selection
advantageous mutations are quickly brought to fixation

most mutations are selectively neutral
have little or no effect on fitness / neither adaptive nor detrimental / not influenced by Darwinian selection
the rate of molecular change of a gene should be regular like a clock

degree of variability of DNA is positively correlated to its rate of evolution

important genes are highly conserved → change slowly. Most of the mutation will be harmful and will be removed
/ less critical genes are poorly conserved → change quickly. Most of the mutations will not be harmful and will accumulate.

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Explain how the substitution of one nucleotide in DNA can be selected for or against in an area endemic with malaria.

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Explain how changes at the genome level can lead to cancer.

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Through genetic drift, these new neutral alleles become more common in the population and become a characteristic feature of the population;

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10 (b) Explain how a point mutation in the gene coding for a growth factor receptor
can lead to cancer. [6]

Mutation in the <u>proto-oncogene</u> that codes the growth factor receptor, results in it becoming an <u>oncogene/ becoming oncogenic</u>

A point mutation results in a change in triplet code/ which changes the amino acid sequence in the growth factor receptor.
The mutated growth factor receptor will have different conformation and function differently.
The mutated receptor might be constitutively activated regardless of the presence of its ligand
OR the mutated receptor might be constitutively active because the mutated receptor does not dissociate from its ligand readily.
This results in <u>persistent intracellular signaling</u> for cell growth and proliferation leading to <u>uncontrolled</u> cell division and formation of tumors
the increase in cell division increased chances of <u>multiple mutations</u> in other critical genes like tumor suppressor genes, DNA repair genes, cell cycle control genes
This results in cells becoming cancerous
Cancerous cells <u>metastasize</u> /spread to other parts of the body to form secondary tumors.
(Cancerous cells can deprive surrounding cells/organs of nutrients and oxygen and cause functional failure of invaded tissues.)

OR

Explain the role of telomeres in maintaining the integrity of a eukaryotic organism. [6]
Are non-coding DNA sequences found at the ends of chromosomes.
Prevents fusion between exposed ends of chromosomes - chromosomal mutation.
Which can disrupt regulatory control of genes on the adjoined chromosomes
Prevents the activation of DNA repair enzymes which will recognize any exposed chromosomal ends as a defect and degrade the chromosomes.
serves as buffer DNA to protect the organism's genes from being eroded after successive rounds of DNA replication
so that critical proteins will still be synthesized in the daughter cells despite the shortened chromosomes
Cell will also undergo apoptosis after a limited number of cell division/mitosis (~50)/when a critical length of the telomere is reached
This limits the extent of accumulated mutations and prevents the development of cancer.

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[1] - 2 sister chromatids per chromosome
[1] – a crossing over of non-sister-chromatids; may also take place on the other arm betw RFLP 4 and 8
[1] – correct/appropriate labeling

E = expected ‘value’

ne;

[1] - 2 sister chromatids per chromosome
[1] – a crossing over of non-sister-chromatids; may also take place on the other arm betw RFLP 4 and 8
[1] – correct/appropriate labeling

RFLP 4

RFLP 8

RFLP 3

RFLP 1

RFLP 8

RFLP 4

RFLP 1

RFLP 3

β-male

RFLP 8

RFLP 1

RFLP 3

RFLP 4

ne

X

nE

RFLP 4

Ne

RFLP 8

RFLP 1

RFLP 3

ne

ne;

nE

Ne

NE

ne;

NE

Y

O = observed ‘value’

c = number of classes

ν = degrees of freedom

$\sum$ = ‘sum of …’

Key to symbols

$\nu = c - 1$

E

$(O - E)^2$

Σ

=

χ^2

χ^2 test

A

B

C

D

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